

Joint biclustering of C1/C2 signal reveals region–pair modules

Biclustering of the joint C1/C2 count matrix revealed a dominant left–right polarization across column (pair-feature) clusters, with signal intensities low on the left and progressively higher toward the right (Figure 1). Row clusters aligned with this gradient: upper rows were uniformly low or mixed-low across columns; several mid-to-lower rows showed sharp, right-restricted enrichment with concomitant depletion on the left. A few narrow row subsets exhibited moderate enrichment limited to middle column clusters, whereas the strongest signals concentrated in bottom-row clusters aligned with the right-hand column groups.

These patterns resolve into rectangular modules—coherent blocks at specific row–column intersections—indicating subsets of genomic regions that co-vary with particular feature groups in the combined dataset. Right-enriched modules show strong, spatially localized activation within right-hand column clusters with minimal spillover leftward; uniformly low blocks delineate a background of inactive regions; and middle-selective modules capture intermediate specificity restricted to a limited subset of columns.

For downstream regulatory interpretation, we aggregated rows by module membership to define region clusters: (i) right-enriched clusters representing high-activity sets specific to right-hand feature groups; (ii) middle-selective clusters with targeted feature preferences; and (iii) uniformly low clusters forming a low-activity background. Because modules are jointly bounded in the row and column dimensions, they capture stable co-variation,

enabling subsequent annotation and enrichment analyses on biologically coherent region clusters anchored to the principal left–right axis of variation in the joint C1/C2 signal.

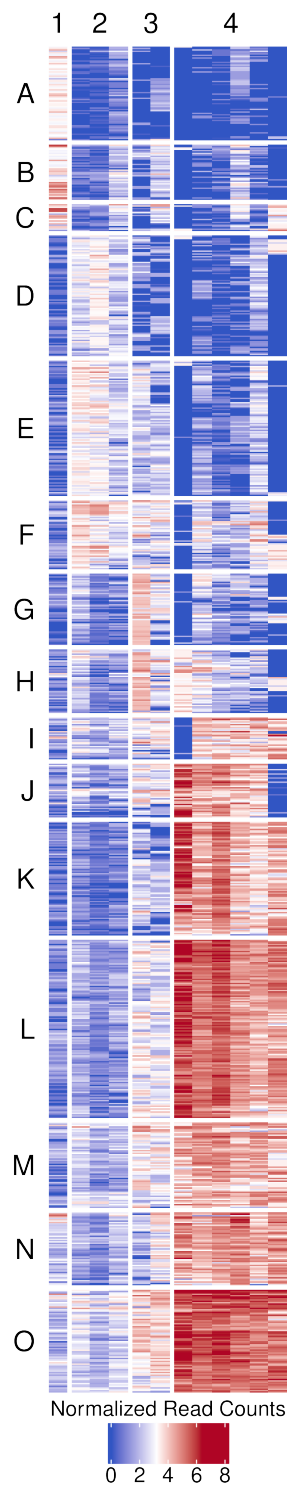


Figure 1. Biclustering reveals co-varying region–pair modules across C1/C2. Rows are genomic regions and columns are pair features jointly clustered from samples C1 and C2. Distinct blocks highlight modules with low signal in left column clusters and high signal in right clusters, indicating coordinated patterns shared across the combined dataset.

Transcription factor motif enrichment stratifies region clusters

Transcription factor (TF) motif enrichment across the bicluster-derived region clusters displayed a strongly polarized landscape (Figure 2). A contiguous block of right-hand clusters showed broadly elevated enrichment for most motifs, consistent with a TF-rich class. In contrast, left-hand clusters exhibited mixed and generally weaker enrichment, and a single early cluster in the left/middle showed a pronounced vertical band of depletion spanning many motifs. Within the otherwise uniformly high right-hand block, sporadic higher-intensity patches indicated additional within-block specificity.

Functionally, the TF-rich right-hand block is consistent with enhancer-like regulation in which multiple TFs act combinatorially; the superimposed patches point to substructure dominated by more selective motif sets that likely refine regulatory outputs within this shared program. The left-hand clusters appear more heterogeneous, with reduced canonical TF binding site density; the conspicuous depletion band marks a cluster selectively lacking common motifs, consistent with a distinct regulatory architecture (e.g., repressed or structural states) relative to the TF-rich block. Because enrichments are computed against an hg38 genomic background, TFs whose motifs peak within each right-hand subcolumn nominate candidate regulators for those region clusters, whereas the relative motif scarcity in the depleted cluster argues against deployment of the same TF programs there.

Together, the joint C1/C2 biclustering stratifies region clusters into at least three regulatory classes—globally TF-rich, heterogeneous/low, and motif-depleted—linking TF co-

enrichment to putative combinatorial control in the TF-rich block and highlighting cluster-specific depletion as a marker of alternate regulatory states (Sousa & Flames, 2022).

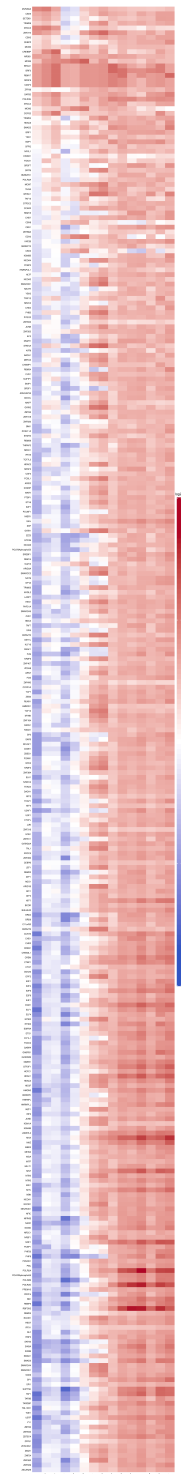


Figure 2. TF motif enrichment across C1/C2 region clusters. Rows are TF motifs and columns are C1/C2-derived region clusters. A broad right-side block shows higher enrichment across most motifs, whereas an early cluster forms a depletion band on the left, indicating cluster-specific motif associations.

Regulatory element classes and genic context of region modules

cCRE class composition and genic context jointly position the region clusters along a continuous distal enhancer–promoter axis. Across clusters A–O, cCRE mixtures resolve from enhancer-like to promoter-like content (Figure 3). Early clusters (A–C) are compositionally mixed, with substantial CA–H3K4me3 together with smaller CA–CTCF/TF components and other accessible states. Mid clusters (D–H) are dominated by distal enhancer-like signatures (dELS) with moderate proximal enhancer-like (pELS) and minimal promoter-like signatures (PLS). Transitional clusters (I–M) show a progressive increase in PLS accompanied by a decline in pELS, culminating in near-exclusive PLS in the terminal clusters (N–O). This progression matches the expectation that distal enhancer classes predominate at gene-distal loci and promoters at transcription start sites (Fulmer et al., 2025; Navratilova et al., 2026) and aligns with the increasing promoter-associated signal intensities that structure the biclustering heatmap.

Genic category distributions mirror these shifts (Figure 4). Early clusters retain larger intergenic and intronic fractions; D–H are intron-heavy with limited promoter overlap; I–K present mixed promoter, 5' UTR, exon, and intron composition with little intergenic content; and L–O are primarily promoter-associated, with promoter and 5' UTR proportions increasing from L to O and peaking in O. Together, these orthogonal annotations delineate a monotonic gradient: clusters enriched for dELS and intergenic/intronic contexts occupy the distal end; transitional clusters (K–L) maintain notable pELS before the PLS inflection in M–O; and the most promoter-like, gene-proximal clusters (N–O) anchor the proximal end. This ordering explains the observed transition in signal structure across the biclustering heatmap and situates each cluster within a functional

continuum from distal enhancer-dense to promoter-enriched regulatory landscapes (Fulmer et al., 2025; Navratilova et al., 2026).

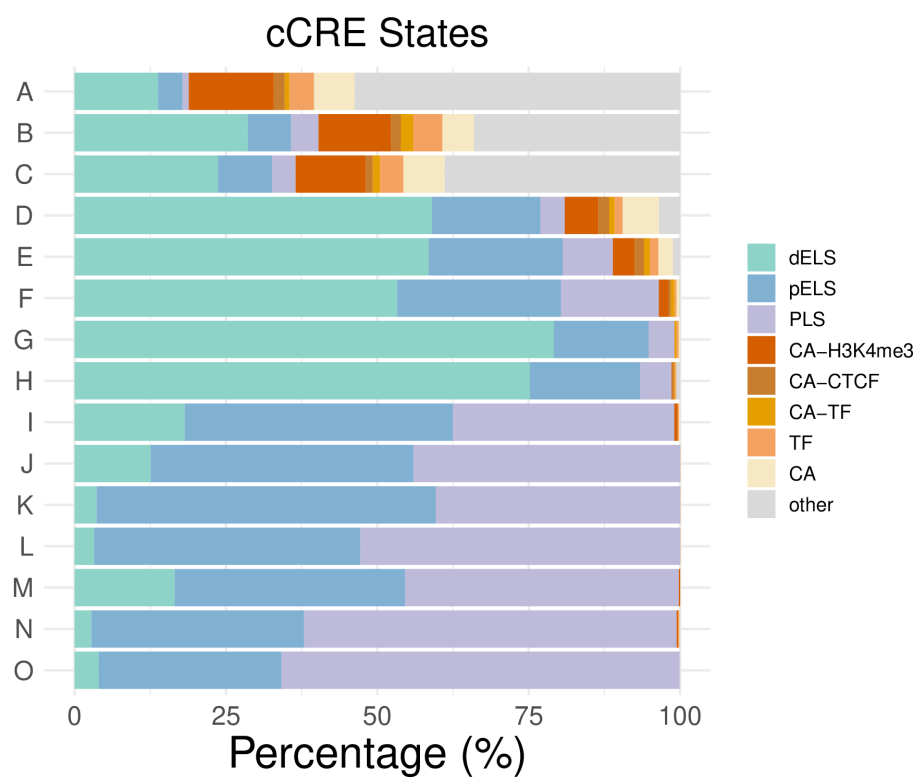


Figure 3. cCRE class composition across C1/C2 biclustered regions. Proportions of cCRE categories are summarized for clusters A–O. Early clusters contain more distal/proximal enhancer states with added CA–H3K4me3 and other classes, whereas promoter-like (PLS) increases along the series and dominates the final clusters.

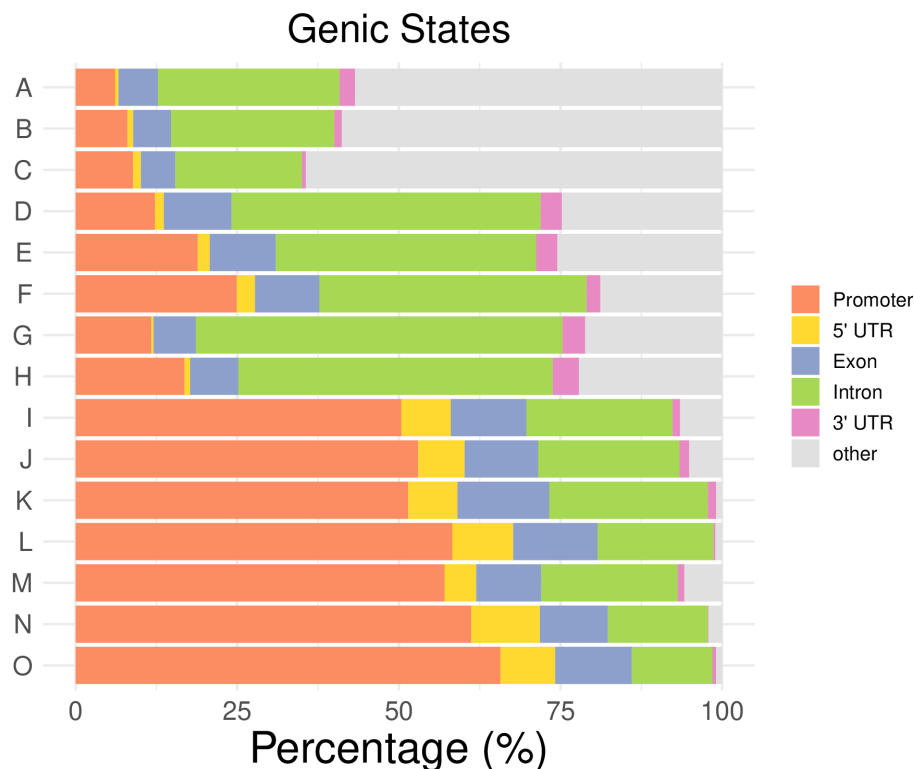


Figure 4. Genic category composition across C1/C2 clusters. Proportions of promoter, 5' UTR, exon, intron, 3' UTR, and intergenic annotations are summarized for clusters A–O. Upper clusters are enriched for intergenic and intronic contexts, whereas lower clusters show increasing promoter and UTR localization, with the promoter fraction highest in O.

References

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